

OPTION PANDA — Scoring System & Strategy Guide

A complete walkthrough of how the dashboard rates every ticker, what each rule measures, and how the strategy presets combine them.

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Executive Summary

OPTION PANDA scores every ticker in its ~2,500-symbol universe against a library of 25+ technical, fundamental, options-flow, and sector rules. Each rule contributes a fixed point weight when its condition fires. The total is clamped to a 0–164 band; tickers at or above the user-tuned Min Score threshold are marked GO.

Five built-in strategy presets layer 'must-pass' filters on top of the score so the user can shift the scanner from broad discovery (any GO) to evidence-gated rarity (HIGH CONVICTION). Each preset is paired with its own minimum score, calibrated against the observed distribution (peaks 140–160 on a ~164-pt practical max as of the 2026-04-30 retune).

The scoring weights are NOT arbitrary. Every rule was derived from a named academic source, an industry-standard methodology, or a council-reviewed backtest. The most heavily-weighted rules (JT 12-1, Minervini Trend Template, Pocket Pivot, U/D Volume Ratio) are the ones with the strongest replicated edge in the literature.

This document covers: the full rule catalog (math + source + weight), the GO-decision logic and ranking tiebreakers, every strategy preset with its filter set and intent, calibration history, and known limitations. A separate companion document covers rule-level and strategy-level impact + efficacy in detail.

1. The Scoring Engine

Every ticker enters `scoreIt()` and runs through ~25 rules grouped into five sections plus overlays. Each rule has a fixed point weight, fires independently, and never short-circuits — a single failing rule doesn't disqualify a ticker, only reduces its total score.

Practical max score = 164 points. Empirical peak in normal markets = 140–160. Default GO threshold = 140.

2. Full Rule Catalog

Every rule the dashboard ships. Weight, what it tests, and the math / academic source. Grouped by section.

Section A — Technical Momentum (Tradier bulk quote)

Rule	Wt	Tests	Math / Source
RSI 40-70	6	Permissive momentum band — not oversold, not extreme.	$RSI(14) \geq 40$ and ≤ 70 .
RSI 50-65 (Cardwell)	6	Cardwell range-shift zone — early entry inside a bull regime.	$RSI(14) \geq 50$ and ≤ 65 . Bonus on top of RSI 40-70.
MACD Hist +	6	MACD histogram > 0 — momentum is positive.	Standard 12/26/9 MACD. $macdHist > 0$.
ADX > 25	6	Confirmed trend strength (Wilder's threshold).	$ADX(14) > 25$.
ADX 15-25 + DI+>DI-	6	Emerging-trend Wilder original signal — DI+ leading DI-.	$ADX(14) \in [15, 25]$ and $DI+ > DI-$. Fires on detail-load only.
EMA20 $>$ EMA50	10	Short-term trend above medium-term trend.	$EMA(20) > EMA(50)$ on daily closes.
TF Aligned 2/3	12	At least 2 of 3 timeframes (daily / hourly / 15-min) point the same direction.	Tradier multi-timeframe quote aggregation. Count of aligned tf ≥ 2 .

Rule	Wt	Tests	Math / Source
Sector PF > 2	5	Ticker's static sector profit-factor > 2 in long-call backtest.	SECTOR_PF lookup table from prior backtests.
52Wk Hi Prox <15%	8	Price within 15% of the 52-week high — universe-pass for breakout setups.	$\text{price} \geq \text{week52hi} \times 0.85$.
52Wk Hi Prox <5% (G-H)	6	George-Hwang JF 2004 effect: returns concentrate within 5% of 52-wk hi.	$\text{price} \geq \text{week52hi} \times 0.95$. Bonus on top of 15% rule.
RS > SPY (21d)	10	21-day relative strength positive vs SPY. Falls back to 5-day if 21d missing.	Total return(21d, ticker) – Total return(21d, SPY) > 0. 5d fallback only when 21d unavailable.
Price > EMA200	6	Above the long-term moving average — long-trend bias.	$\text{price} > \text{EMA}(200)$. Falls back to $\text{EMA}20 > \text{EMA}50$ if EMA200 missing.

Section B — Fundamentals / FMP

Rule	Wt	Tests	Math / Source
Analyst Revisions ↑	12	≥2 analysts have raised estimates and there are recent upgrades.	FMP analyst estimates: revisionsUp ≥ 2 AND upgradesLast30 > 0. Long-calls bias: upward revisions outperform downward revisions.
Analyst PT Exists	8	A consensus analyst price target is published.	FMP price target: ptConsensus present. Light fundamental confirmation.

Section C — GEX / Options Flow

Rule	Wt	Tests	Math / Source
GEX Call Flow (55–69%)	4	Emerging call-side OI bias in the 30-DTE chain (constructive but not crowded).	GEX call% of OI ∈ [55, 69].
GEX Call Flow (70–74%)	6	Strong but not overcrowded call-side flow.	GEX call% of OI ∈ [70, 74].
GEX Call Flow (≥75%)	-6	Overcrowded call-side OI — contrarian signal, dealers pin/fade.	GEX call% of OI ≥ 75. PENALTY tier.

Section D — Momentum Acceleration (history-based)

Rule	Wt	Tests	Math / Source
MFI(14) > 50	10	Money Flow Index above midline — rising volume-weighted momentum.	$\text{MFI}(14) > 50$ (RSI mid-line analog with volume).
CMO(9) > 50	7	Chande Momentum Oscillator > 50 (Chande's original 1993 threshold).	$\text{CMO}(9) > 50$.

Section E — Evidence-Based Swing Rules (history-based)

Rule	Wt	Tests	Math / Source
Minervini Trend Template	15	Stage 2 trend confirmation — 6 of 8 SEPA criteria computable from daily bars.	All six must pass: price > MA50; price > MA150; price > MA200; MA50 > MA150 > MA200; MA200 rising vs 22 bars ago; ≥30% above 52-wk low.
Pocket Pivot	15	O'Neil / Morales-Kacher institutional accumulation signal.	Today's volume > largest down-day volume of prior 10 days AND close > MA10 AND close in upper half of day's range AND ≤5% extended from MA10.
U/D Volume Ratio (50d)	+12 / +8 / +4	Up-volume vs down-volume ratio over trailing 50 sessions (O'Neil IBD).	$\Sigma(\text{volume on up-close days}) / \Sigma(\text{volume on down-close days})$ over 50d. $\geq 2.0 = +12$ strong; $\geq 1.5 = +8$ accumulation; $\geq 1.25 = +4$ mild; $< 1.25 = 0$.
VCP — ATR Contraction	12	Volatility Contraction Pattern: supply drying up before breakout.	$\text{ATR}(14) \text{ today} / \text{ATR}(14) \text{ 30 sessions ago} < 0.70$. Minervini methodology.
BBW Squeeze (<p20 of 120d)	10	Bollinger Band-width compressed in the bottom 20% of its trailing range.	$\text{BB}(20, 2\sigma) \text{ width} / \text{mean}$. Today's value ≤ 20th-percentile of last 120 days.
20d Donchian Breakout + Vol	12	Turtle Traders breakout — Dennis / Eckhardt 1983-88 methodology.	today close > 20-day high AND today volume ≥ 1.5 × 50-day avg volume.
JT 12-1 Momentum	20	Cross-sectional momentum, skip last month (Jegadeesh-Titman 1993).	Return from t-252 to t-21. Threshold: ≥ +25% (top-quintile proxy). Single strongest replicated factor in equity literature.

Overlays & Penalties

Rule	Wt	Tests	Math / Source
PROVEN ticker	8	Ticker is on the curated PROVEN good-mover list.	Hand-maintained list. +8 flat bonus.
AVOID ticker	-15	Ticker is on the curated AVOID list (chronic chop / scams / fundamentals).	Hand-maintained list. -15 flat penalty and forced NO-GO.
Strong Sector	+8 / -8	Daily + intraday theme RS overlay vs SPY.	Combined daily + intraday theme score. $\geq 75 = +8$; $\leq 25 = -8$.
Broken Trend (penalty)	-5	Price below EMA50 — medium-term trend broken; calls fight gravity.	price < EMA(50). Penalty only — no positive credit.

3. GO Decision & Ranking

How the GO flag is set

After all rules fire, the raw score is clamped to the 0–164 band. Penalties (Broken Trend –5, weak sector –8, AVOID –15) can drive the subtotal negative; the clamp forces it to 0.

GO threshold is read from the Min Score input on the scanner tab. Default 140 was calibrated 2026-04-30 against the observed distribution (peaks 140–160 on a ~164-pt practical max). Strategy presets override the default with their own per-strategy min-score.

AVOID-list tickers can NEVER be GO regardless of score.

Tiebreaker order

Priority	Field
1. Score	Primary sort — descending.
2. techScore	Sum of Section A + D + E rules.
3. relStr5d	5-day relative-strength % vs SPY.
4. Theme dailyScore	Sector / theme strength from theme_scores.json.
5. Volume	Daily volume — liquidity floor.

4. Strategy Presets

Five preset combinations layer must-pass filters on top of the score, plus their own minimum score threshold.

★ HIGH CONVICTION — Min Score 140

Intent. Rarest, highest-quality setups. Stacks the two highest-IC scoring rules (JT 12-1 at +20 and Minervini at +15) as MUST-PASS instead of merely contributing to the score.

Must-pass filters.

- PROVEN ticker
- Analyst Revisions ↑
- GEX Call Heavy ($\geq 55\%$)
- JT 12-1 Momentum ($\geq 25\%$, top-quintile proxy)
- Minervini Trend Template (6/6)

Expected matches. 0–5 matches per scan in normal market conditions.

▲ BREAKOUT — Min Score 125

Intent. Chart-pattern-driven breakout buying with momentum + leadership. Donchian and VCP are the two highest-IC pure-pattern breakout rules; requiring BOTH filters out fake-out names.

Must-pass filters.

- 52Wk Hi Prox $< 15\%$
- RS > SPY
- MACD Hist +
- TF Aligned 2/3
- 20d Donchian Breakout + Vol
- VCP — ATR Contraction

Expected matches. 5–25 matches per scan, market-regime dependent.

■ MOMENTUM — Min Score 120

Intent. Confirmed trending stocks with sector tailwind, gated by the academic gold-standard 12-1 cross-sectional momentum signal.

Must-pass filters.

- MACD Hist +
- Price > EMA200
- RS > SPY
- Strong Sector
- JT 12-1 Momentum

Expected matches. 5–30 matches per scan during trending markets.

◆ BEST ITM CALLS — Min Score 145

Intent. Deep-ITM (δ 0.70–0.90) calls on names that pass ALL THREE of the highest-IC scoring rules. You're paying real premium for deep-ITM contracts — only setups confirmed by 12-month momentum + Stage 2 trend template + institutional accumulation should graduate.

Must-pass filters.

- MACD Hist +
- Price > EMA200
- RS > SPY
- JT 12-1 Momentum
- Minervini Trend Template
- Pocket Pivot
- Price $\geq$ \$5
- Avg Volume $\geq$ 1,000,000
- Vol ratio $\geq$ 100% (today vs avg)
- Max option price $\leq$ \$8

Expected matches. 0–8 matches per scan; very selective.

▲ PRE/POST MARKET — Min Score 140

Intent. Extended-hours gappers screened with the SAME filters as HIGH CONVICTION, then re-sorted by absolute gap %. Use any time the extended-hours session is active or on weekends.

Must-pass filters.

- PROVEN ticker
- Analyst Revisions $\uparrow$
- GEX Call Heavy ($\geq 55\%$)
- JT 12-1 Momentum
- Minervini Trend Template
- Sort: |gap %| descending (overrides standard score sort)

Expected matches. 0–5 matches.

5. Calibration History

Dated record of every retune.

2026-04-29 council overhaul (drops)

Removed three anti-signal rules: RSI(7) Accel, 5d %chg Accel, Vol > Avg flat threshold.

2026-04-29 council overhaul (retunes)

Tightened existing rules to evidence-based parameters: Cardwell range-shift, MFI(14)>50, CMO(9)>50, 21d RS, George-Hwang 52-wk bonus.

2026-04-29 council overhaul (additions)

Added 7 evidence-based rules: Minervini, Pocket Pivot, U/D Vol Ratio, VCP, BBW Squeeze, Donchian, JT 12-1. Practical max raised from ~110 to ~164.

2026-04-30 GO threshold retune

Default GO threshold 100 → 150 → 140. Distribution peak sits 140–160, not 165+.

2026-05-01 GEX retune

GEX Call Flow moved to tiered +4/+6/-6 with extreme-positioning penalty above 75%.

2026-05-04 Analyst + sector bumps

Analyst Revisions +12 → +16 effective weight; Strong Sector $\pm 5 \rightarrow \pm 8$ — long-calls bias depends more heavily on bullish signals and sector tailwind.

6. Known Limitations

Honest framing for new users.

- Single-leg long-call bias. The scoring is tuned for long-call swing trades, NOT validated for short premium, spreads, puts, or longer-than-50-DTE positions.
- Backtest $n=19$ trades (2026-04-29 council review). Empirical edge of the combined system is small and not statistically separated from noise. Treat rule weights as expert-judgment-with-priors, not empirically-validated weights.
- Some rules require ≥ 210 daily bars (Minervini, Donchian, U/D Volume, BBW). Fresh IPOs are flagged `dataGap=true` and treated as 'unknown', NOT failed.
- GEX data is sourced separately (runs once per scan + scheduled 9:35/12:00/15:00 ET). If GEX hasn't run yet the GEX rule scores 0 — neither bonus nor penalty.
- Cross-sectional ranking (JT 12-1's true form) is approximated via a hard +25% threshold rather than ranking against the universe simultaneously.
- Theme overlay depends on `theme_scores.json` regenerated by a nightly GitHub Actions cron. If the cron fails, the Strong Sector rule pass-throughs.

7. End-User Playbook

1. Set your API keys

Open the API tab and enter your Tradier proxy URL + live-mode token, and your FMP API key. Without an FMP key, Section B rules score 0 — the rest still works.

2. Pick a strategy preset

Tap STRATEGY at the top of the scanner. Start with HIGH CONVICTION to see the most selective gate, or BREAKOUT / MOMENTUM for more candidates. CLEAR returns to default.

3. Run the scan

Tap ■ SCAN. Bulk pass scores ~2,500 tickers in 30–60s. Foreground enrichment adds Tradier history + FMP data to the top 60. Background pass extends Tradier history to ranks 61–200 over ~5 minutes.

4. Read the results table

Click any row to open the detail pane with the full rule-by-rule breakdown — every passing rule with its point value, every failing rule with what it would have scored.

5. Use Export Matrix or Export GOs

EXPORT GOs copies a comma-separated list of every GO ticker. EXPORT MATRIX in the Portfolio tab pulls all positions into a Google Sheet.

6. Tune Min Score if needed

If a preset returns 0 matches in a quiet market, drop Min Score by 10–20. If 50+ in a hot market, raise it to tighten.

Appendix — Glossary

Term	Definition
GO / NO-GO	Binary classification. GO = score $\geq$ Min Score AND not AVOID-listed.
Min Score	User-tunable input (default 140). Strategy presets override.
Must-Pass	Filter applied AFTER scoring. Failing a must-pass excludes the ticker.
techScore	Sum of Section A + D + E rule weights. Second tiebreaker.
fundScore	Sum of Section B + Strong Sector. Analyst + sector contribution.
GEX	Gamma exposure — dealer-hedging pressure from the 30-DTE chain. Call Heavy = $\geq 55\%$ of OI on calls.
JT 12-1	Jegadeesh-Titman 12-1 momentum: return from t-252 to t-21.
VCP	Volatility Contraction Pattern — Minervini's ATR-shrinkage precursor.
Donchian	Turtle Traders breakout — close above prior 20-day high on 1.5 $\times$ + volume.
BBW	Bollinger Bandwidth — bottom 20% of trailing 120-day range = squeeze.
PROVEN ticker	Curated list with historically favorable long-call P&L. +8 bonus.
AVOID ticker	Curated AVOID list. -15 + forced NO-GO.
Theme overlay	Sector / theme RS from theme_scores.json (nightly cron).
Intraday RS	Live theme RS computed from member changePct vs SPY during the day.

Companion document: [OPTION_PANDA_Impact_Efficacy_Report.pdf](#) covers rule-level + strategy-level efficacy in detail.